Technical Appendix

Measuring the Direction of Technological Change with a Language-Model Research Loop

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October 2026

1 Scope

This appendix gives the construction and the validation of the three steps of the loop described in the proposal, at the level of detail needed to evaluate them. Sections 3 to 5 and Section 6 reproduce the corresponding sections of the companion paper (Rony, Ciarli, and Nomaler, 2026), which reports the full results. Section 2 summarises the first step, the construction of the technology taxonomy, whose full account, with the coverage diagnostics, the configuration search and the validation of the clustering, is Sections 2 and 3 and Appendix B of the companion paper. The calibration of the rating instrument against a panel of some 600 domain experts is the subject of a second companion paper (Rony and Ciarli, 2026). Every data file, verdict and script named below is in the repository that accompanies the proposal.

2 The taxonomy in brief

The digital technologies are defined by a taxonomy built from Wikipedia and patents. Starting from 27 curated seed categories, a breadth-first crawl of the English Wikipedia category graph to a depth of four levels yields about 1.8 million page records. A five-stage cleaning cascade reduces them to about 78,000 pages, whose descriptions are matched against 1.9 million digital patent abstracts in the embedding space of PatentSBERTa, a sentence-embedding model trained on patent text (Bekamiri, Hain, and Jurowetzki, 2024). Noise filters on titles and a patent-similarity threshold of 0.6 leave 11,906 pages, and a language model acting as judge (Gilardi, Alizadeh, and Kubli, 2023; Gu, Jiang, Shi et al., 2024) classifies whether each remaining page describes an actual technology and accepts 8,069. Table 1 reports the funnel.

The accepted pages are grouped by community detection on the wiki-link graph, with every edge weighted by the semantic similarity of the two pages it joins. The Leiden algorithm (Traag, Waltman, and van Eck, 2019) partitions the graph from the top, first into root domains, then each root into intermediate clusters and each of these into base technologies, with the resolution chosen separately for every group that is split so that the number and size of its children suit the level. The free parameters were selected by a grid search on the within-minus-between similarity margin at the worst of the three levels. The raw clustering, 12 root, 56 intermediate and 205 base clusters, becomes the final taxonomy in two stages. A correction ledger records some 1,800 structural and page-level corrections in a first pass and, in a second, the split of 30 base clusters into 62 and some

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1,100 page moves. A language-model review then reads every technology with its pages and judges, on technology content alone, whether each page belongs to its technology, to another technology or to none. Its proposals were adopted after inspection, in two rounds, and every move, removal and new technology is recorded with its item identifier. The final taxonomy has 10 root domains, 43 intermediate and 184 base technologies on 6,928 pages (Table 2).

The taxonomy is validated with information its construction never uses. About 70 percent of the pages lie closer to the centre of their own base technology than to any other. In the median base technology, patents of the same technology share patent classes more often, and cite one another about thirty times as often, as patents of different technologies. The candidate pages match 94 percent of the digital patent corpus, and the unmatched patents form no large citation-connected group. For the green domain the loop adopts the OECD ENV-TECH classification (Haščič and Migotto, 2015), whose 66 technology nodes in ten groups are defined by mappings from the patent classification codes, so that a patent's membership follows from its codes alone. The two domains therefore enter the exposure measure through different attribution rules, by text against pages and by classification codes, and the difference shows in the share of cells the judge later sets to zero (Section 3.2).

Table 1: Construction funnel of the digital taxonomy, from seed categories to the final taxonomy.

Stage	Object	Count
Seeds	Curated Wikipedia categories	27
Crawl	Raw (page, category-path) records	1,801,777
Cleaning	Pages surviving the cleaning cascade	78,049
Matching	Pages embedded and matched against patents	70,714
Filtering	Pages after the noise filters and the 0.6 threshold	11,906
Judging	Pages accepted by the LLM as judge	8,069
Clustering input	Pages after removing listed noise titles	8,011
Corrections	Pages after the correction ledger	7,398
Review, first pass	Pages after the page review	7,351
Review, second pass	Pages in the final taxonomy	6,928
Final taxonomy	Root / intermediate / base technologies	10 / 43 / 184

3 The exposure measure

Exposure measures how strongly each technology bears on the work of each occupation and the activities of each industry. The measure reads this from text, comparing what the technology's patents describe with what the occupation's tasks or the industry's activities describe, so that it can be computed identically for every technology of both domains.

3.1 Measuring exposure

For a technology n and an entity e, an occupation or an industry, exposure measures technological relevance, the degree to which the content of the technology's patents resembles the tasks of the occupation or the activities of the industry. The measure is neutral with respect to direction by design. The bank-teller example that opens the companion paper shows that the sign of a technology's employment effect is not a property of the technology alone, so exposure delivers the magnitude of contact and leaves the direction to the separately rated measure of Section 4. Occupations are

Table 2: Root domains of the final digital taxonomy.

Root domain	Intermediate clusters	Base technologies	Pages
Computing Hardware and Systems	3	15	1,333
Software and Data Management	8	45	1,136
Networking and Telecommunications	4	12	954
Signal Processing and Communications	7	32	914
Artificial Intelligence	4	10	791
Imaging and Display Technology	2	13	566
Cryptography and Security	2	11	474
Robotics and Manufacturing	5	24	455
Transport and Navigation Systems	3	6	201
Quantum Computing	5	16	104
Total	43	184	6,928

427 of the 436 four-digit unit groups of the International Standard Classification of Occupations (ISCO-08), represented by some 2,800 task sentences from the European Skills, Competences, Qualifications and Occupations classification (ESCO), and industries are the 272 three-digit groups of the Statistical Classification of Economic Activities in the European Community (NACE Rev 2), represented by some 850 activity sentences (Appendix A of the companion paper).

The construction builds on Prytkova, Petit, Li, Chaturvedi, and Ciarli (2024), who relate patent and task text, and connects three corpora, the entities' task or activity sentences, the patents and the taxonomy pages, through three similarity relations with distinct roles. Membership determines which patents constitute a technology, value measures how strongly a patent resembles an entity's tasks or activities, and gating determines whether a sentence relates to the technology's subject at all. Because the three relations are separate, their errors can be corrected separately (Section 3.2).

Membership operates in the PatentSBERTa space in which pages and patents were matched for the taxonomy. Each patent is tagged to the pages among its twenty most similar with a cosine similarity of at least 0.40, and it counts towards a single technology, the one that contains its best-matching page. Under a many-to-many rule, in which a patent joins every technology in which it has a qualifying page, a small number of broad technologies absorbs a disproportionate share of the corpus. The value of a patent for an entity sentence is their cosine similarity in the space of the general-purpose sentence-embedding model all-mpnet-base-v2 (MPNet below) (Song, Tan, Qin, Lu, and Liu, 2020; Reimers and Gurevych, 2019), counted only above a floor of 0.32, the 99th percentile of the sentence-patent similarities. The floor is defined on the percentile scale because similarity scales differ by about a factor of two between the two embedding models and differ across the three relations within a model, so that an absolute threshold selects tails of very different depth (Appendix C.3 of the companion paper). A sentence contributes to a technology through a patent only if its PatentSBERTa similarity to the page through which the patent belongs to the technology reaches 0.40. This page gate prevents value from accruing through patents whose page is unrelated to the entity's work. Each patent may give value to at most the ten entities whose sentences resemble it most once each entity's baseline similarity to all patents is subtracted, a variant of the top-ten relevance filter of Prytkova et al. (2024), so that no single generic patent contributes to every entity.

Figure 1 reports, over some 19 million random triples of a task, a patent and a page, the mean task-page similarity within bins of the task-patent and the patent-page similarity. It is highest where both are strong. Part of this agreement is geometric, since a task close to a patent that is

close to a page is itself partly close to the page. The figure establishes that the three relations are consistent, a precondition for requiring all three jointly. The joint requirement removes links that are strong in one relation only, such as a patent that resembles a task through shared vocabulary while belonging to an unrelated page.

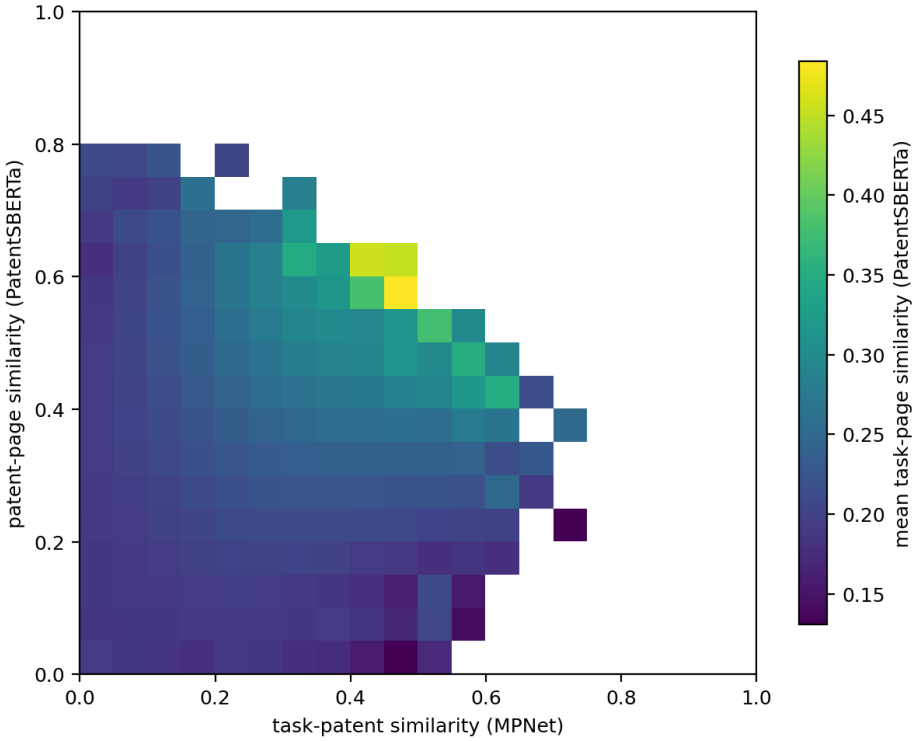


Figure 1: Agreement of the three similarity relations of the exposure measure. The sample is 20 million randomly drawn triples of a task sentence, a patent and a page of the taxonomy before the language-model review, of which the 18.7 million with non-negative task-patent and patent-page similarity are binned. Each cell reports the mean task-page similarity (PatentSBERTa, the gating relation) within a bin of the task-patent similarity (MPNet, the value relation, horizontal) and the patent-page similarity (PatentSBERTa, the membership relation, vertical). Empty cells hold no triples. The task-page similarity is highest where the other two relations are jointly strong, 0.37 in the bins in which both exceed 0.5 against 0.20 over all triples.

Let $S(e)$ denote entity e's sentences, $P_n(w)$ the member patents of technology n filed in window w, ν_p the citation weight of patent p, its forward citations relative to the mean of its filing-year cohort, so that an uncited patent carries no weight, $A(s,p)$ the value-relation cosine similarity, and $g(s,p)$ the indicator that the pair passes the floor, the page gate and the relevance filter. The exposure of entity e to technology n over window w is

$$x_{en}(w) \;=\; \operatorname{asinh}\left(\frac{1}{|S(e)|} \sum_{s \in S(e)} \sum_{p \in P_n(w)} \nu_p \cdot \max\{0,\, A(s,p)\} \cdot g(s,p) \right), \tag{1}$$

the mean over the entity's sentences of the citation-weighted sum over the technology's patents, rescaled once by the inverse hyperbolic sine. The mean over sentences rewards technologies whose patents relate to many of the entity's tasks, while a maximum would reward a single task. Ap-

pendix C.4 of the companion paper lists the alternative aggregations. Within a sentence the sum runs over all member patents that pass the gates, so exposure grows with the number of such patents. The measure is computed year by year, with each patent weighted by its citations relative to the mean of its filing-year cohort (Hall, Jaffe, and Trajtenberg, 2001), and one untransformed value is stored per year, so that exposure over any window is the inverse hyperbolic sine of the summed yearly values and the nonlinearity is applied exactly once. An entity's exposure to a domain is the mean over the domain's N technologies, $X_e(w) = N^{-1} \sum_n x_{en}(w)$, with technologies that do not reach the entity counted as zero. Exposure to a digital root or intermediate technology, or to a green group, is the mean of $x_{en}(w)$ over the base technologies it contains, with zeros included, the rule of the domain average X_e applied within the root or group. The green measure follows the same construction except in membership, which the ENV-TECH indicators fix and which needs no page gate (Appendix C.1 of the companion paper).

3.2 The LLM as judge

The last step of the construction uses the LLM as judge. It reads the entity rankings and the largest patent links, the contributions of single patents to an entity's exposure, and sets to zero those without technological content. It applies two criteria. The top-ranked entities of a technology must be plausible users or producers of it, and no ranking may be driven by a lexical coincidence. In the digital domain it reviews up to the fifty largest patent links of every exposed technology, for occupations and for industries. Its verdicts on some 15,000 links move almost 4,000 patents to the technology to which they belong, remove about 300 that are not digital technologies and set some 6,400 links to zero. It also sets to zero the entity-technology cells driven by a lexical coincidence, such as refuse sorters ranked on a data-structures node through the word sorting, 1.5 percent of the exposed digital cells. In the green domain, where membership is fixed by classification codes, it sets to zero 0.07 percent of the exposed cells, those of two occupations of drivers of animal-drawn and hand-propelled vehicles and one industry, the manufacture of other transport equipment, ranked on the transport nodes through the word vehicle. No technology is dropped for lack of exposure. The digital and the green measures share the value relation, the entity text and the aggregation. They differ in the patent corpus and in how patents are attributed to technologies, through their text against pages with a page gate or through their classification codes, so the difference in the share of cells set to zero arises in the text-based attribution, although only the digital measure also passes through the link review (Appendix C.2 of the companion paper).

3.3 Validity of the exposure measure

For external validity, the occupation-side exposure profiles are compared with ten published exposure indices over 356 four-digit ISCO occupations, with exposure over 2009–2023. The ten indices are the AI occupational exposure index of Felten, Raj, and Seamans (2021), the model- and human-rated language-model exposure of Eloundou, Manning, Mishkin, and Rock (2024), the generative-AI exposure of Eisfeldt, Schubert, and Zhang (2023), the AI applicability index of Tomlinson, Jaffe, Wang, Counts, and Suri (2025), the computerisation probability of Frey and Osborne (2017), the AI, software and robot indices of Webb (2020), and the aggregate technology exposure of Prytkova et al. (2024). Figure 2 reports the Spearman rank correlation of each of our ten root-level profiles with each index and leaves insignificant correlations blank. The Artificial Intelligence root correlates positively with all five AI and generative-AI indices. The Robotics and Manufacturing root shows the opposite pattern, correlating positively with the Webb robot and software indices and with Frey and Osborne (2017) and negatively with the five AI and generative-AI indices. The Transport

and Navigation Systems root, which the review of the taxonomy separated from the robotics root, follows the robotics pattern, negative with the five AI and generative-AI indices and positive with the Webb robot, software and AI indices. The broad aggregate of Prytkova et al. (2024) correlates positively with every root but quantum computing, whose occupation profile correlates weakly and positively with four of the five AI and generative-AI indices, at 0.10 to 0.14, and with none of the other indices. The aggregate of Prytkova et al. (2024) shares the approach of our measure and one of its authors, so its correlation with the roots is partly mechanical, and the other nine indices carry the comparison. The AI index of Webb (2020) correlates only weakly with the Artificial Intelligence root, at 0.12. It correlates more strongly with the roots of physical and signal technologies, robotics and manufacturing, transport and navigation, computing hardware and signal processing, as does Webb's software index, and Webb's robot index with the first three. A size or generic-technology artefact would correlate every root with every index. The sign reversals argue against such an artefact.

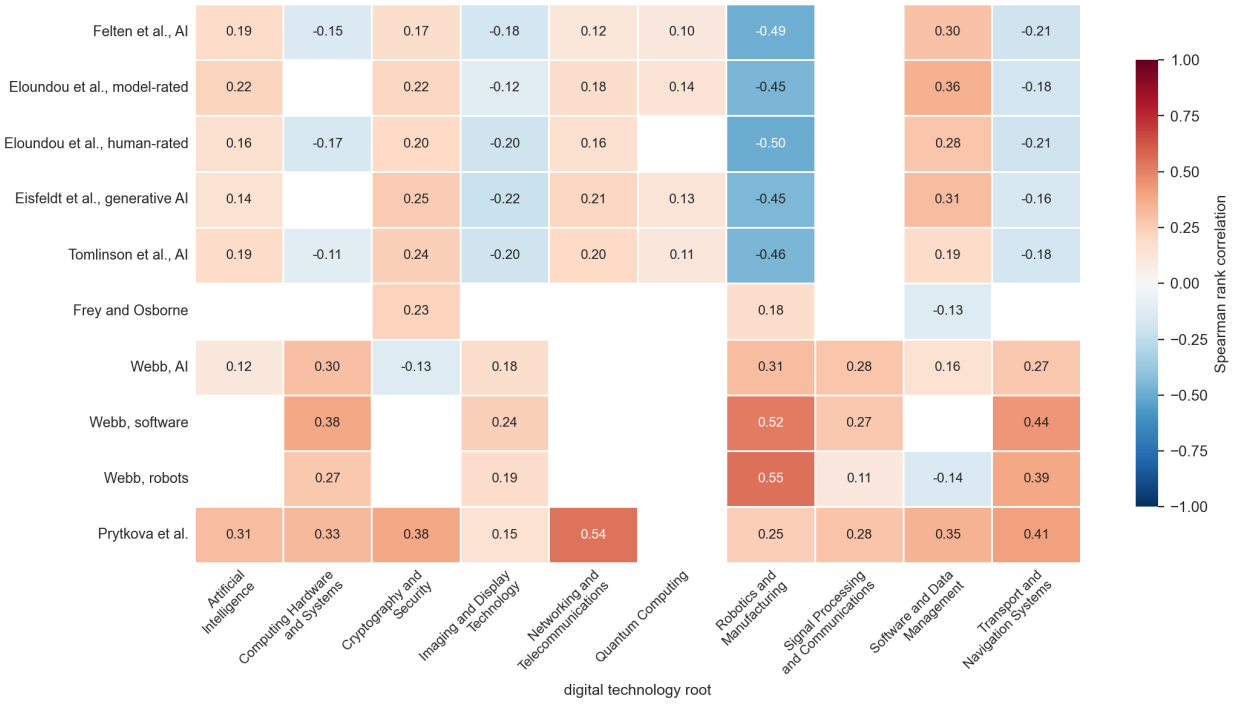


Figure 2: External validity of the occupation exposure profiles of the roots. The cells report Spearman rank correlations with ten published exposure indices over 356 four-digit ISCO occupations, with exposure over 2009–2023. Blank cells are insignificant at the five-percent level. The Artificial Intelligence root correlates positively with the five AI and generative-AI indices, and the Robotics and Manufacturing and the Transport and Navigation Systems roots correlate negatively with them and positively with the robot index. The aggregate index of Prytkova et al. (2024) correlates with every root except quantum computing.

Internally, the measure rests on the agreement of its three relations (Figure 1), and the measure is validated by reading its complete entity rankings, because the relevant failure modes, lexical coincidences and concentration artefacts, are not visible in correlation-based statistics. Figure 3 reports the most exposed occupations by domain. The top digital occupations are information and communication technology (ICT) managers, ICT operations and support technicians and applica-

tions programmers, together with precision-instrument makers, photographers and medical-imaging technicians. The top green occupations are chemical, power and steam plant operators, incinerator and water-treatment plant operators, air-conditioning and refrigeration mechanics, and refining operators. At the level of individual technologies, for example, Electronic Payment Processing ranks bank tellers and cashiers among occupations and monetary intermediation among industries, and Medical Imaging ranks medical-imaging technicians and medical-device manufacturing. Geothermal energy ranks well drillers first, and wastewater treatment ranks water-treatment plant operators first. *The chemical and refining operators at the top of the green ranking are rarely counted among green occupations, and the technologies behind their rank show why. Chemical processing plant controllers, the most exposed occupation, rank first among occupations on the technologies of oil refining and petrochemicals, on fuel cells, on hydrogen for transport and on environmental monitoring, and second on the technologies of the chemical industry, on hydrogen technology and on the capture of carbon dioxide and of methane. Petroleum and natural gas refining plant operators rank second on oil refining and petrochemicals and on hydrogen for transport and third on fuel cells and on hydrogen technology. These ENV-TECH classes describe cleaner production processes and new fuels in the industries with the largest emissions, and the occupations that run those processes are the ones the measure places at the top. Environmental innovation is most relevant to the activities with the largest environmental impact, and the ranking reflects that.*

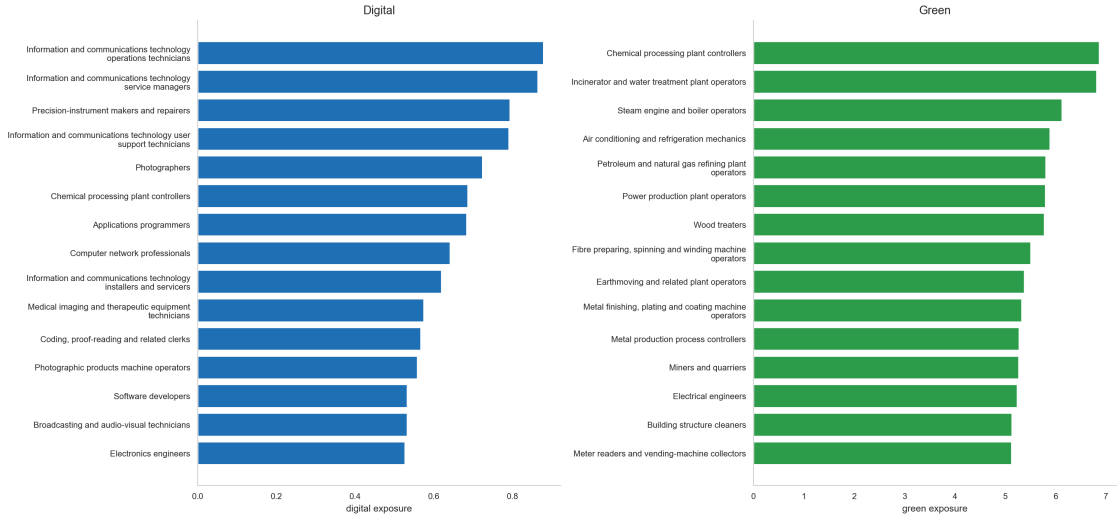


Figure 3: The most exposed occupations by domain (exposure over 2009–2023). Digital exposure is led by ICT occupations together with precision-instrument and imaging occupations, and green exposure by plant and machine operators in energy, water and process industries.

The residual occupation-side limitation is a bias towards concretely worded task statements. Task text written in operational language is systematically closer to patent text than abstract task text, so that under some aggregations abstract technologies rank manual occupations above their actual users. Appendix C.5 of the companion paper describes the bias.

4 The substitution and complementarity measure

The direction measure rests on two constructs, substitution and complementarity, rated by a large language model under workplace personas, roles that the model is asked to adopt, for the technology-

occupation-industry cells in which exposure concentrates (Section 6).

4.1 Unit and rating design

The rated unit is a triplet consisting of a technology node (a digital base technology or a green ENV-TECH node), a three-digit ISCO occupation group, and a NACE Rev 2 section, the finest occupation-industry resolution at which European employment data record the pair. The base-technology level is chosen because we expect specific technology labels to draw specific assessments, and ratings can be aggregated to coarser levels while the converse is not possible.

Two constructs are rated. Substitution is defined as the technology taking over the work itself, so that fewer workers are needed, and complementarity as the technology supporting or enhancing the workers, who continue to perform the work at higher quality or speed. Each construct is presented as an agreement statement and rated on a scale from 1 (strongly disagree) to 5 (fully agree), with 0 defined as do not know. The statements name no time horizon, because the technologies have patents filed within the exposure window 2008 to 2023. Section 6.1 gives the prompts verbatim. The instrument has no separate not-applicable option, and Section 6.5 shows that the zero is used as one. Each statement names the rated object as a digital or a green technology according to the domain of its node, and this domain label supplies the context in which the node's name is read.

Each rating is given under one of three personas, specified by a single role sentence. They are a manager who runs the team performing the work, an experienced worker who performs it, and a chief executive responsible for the organisation's strategy and investment decisions. Ratings are stored per persona. The chief executive represents the side of the employment relation that decides on adoption, since investment and staffing decisions are the channel through which an assessment of substitution reaches labour demand.

The respondent model is DeepSeek-V3.1, a model with open weights, queried as `deepseek/deepseek-chat-v3.1` through the OpenRouter interface in September and October 2026, at temperature 0.3 and top-p 0.3, settings that limit the randomness of the sampled answers, as selected in our companion work on language-model survey panels (Rony and Ciarli, 2026). Responses are constrained to a fixed machine-readable format, a JSON object holding one integer rating followed by a short justification. Each triplet receives 30 ratings, across two constructs, three personas and five independent draws. Each call, a single query to the model, carries one construct and one persona. In a test on an earlier version of the prompt, merging constructs or personas into one call moved the ratings away from those of separate calls, reduced the use of the zero category and raised the draw-to-draw noise (Section 6.2). Scale, response format and the one-construct, one-persona call follow Rony and Ciarli (2026).

4.2 The rated grid

The grid, the set of triplets put to the respondent model, is restricted to cells in which both exposure and employment indicate that the question is meaningful. It uses the exposure of Section 3, computed over the window 2008 to 2023 and aggregated from four-digit occupations to three-digit groups and from three-digit industries to sections by unweighted means over the exposed codes. A triplet qualifies if the exposure of its occupation group to the node and the exposure of its section to the node both exceed a threshold. The thresholds are common to all technologies, with one for occupations and one for industries in each domain. A node-specific threshold would admit the upper tail of nodes whose exposure is low throughout. Each threshold is the 70th percentile of the corresponding distribution of exposed cells (Section 6.3 gives the values). While the grid is selected on exposure over 2008 to 2023, the descriptive statistics use exposure over 2009 to 2023, and the

employment test builds the contents from the patents filed in the years of each outcome period.

The employment filter retains occupation-industry pairs accounting for at least 0.1 percent of the occupation's pooled 2011–2023 employment in the LFS, a floor that removes pairs with negligible employment. The grid has some 12,000 triplets, a third digital and two thirds green, on 112 digital and 62 green technologies. Appendix A of the companion paper reports its dimensions. Entities enter the prompt under their names and green nodes under their ENV-TECH descriptions, since a bare classification code does not tell the respondent what the technology is.

4.3 The ratings

The runs yield 30 usable ratings for every triplet, some 360,000 in all. The few calls that failed or returned no rating were rerun. Table 3 and Figure 4 report their distribution.

Table 3: Ratings by domain and construct. The scale runs from 0 to 5, and the zero category is counted as its bottom.

Domain	Construct	Mean	SD	n
Digital	Complementarity	3.472	1.178	60,720
Digital	Substitution	2.223	0.869	60,720
Green	Complementarity	2.965	1.357	118,935
Green	Substitution	1.934	0.911	118,935

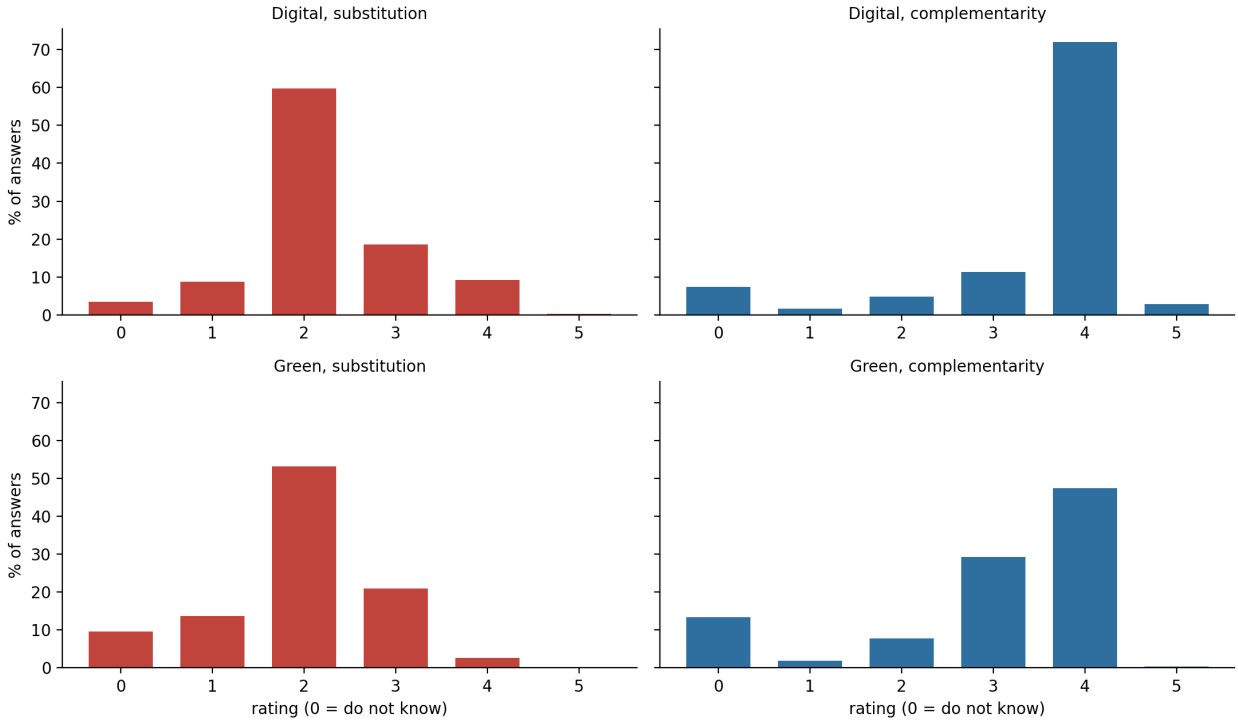


Figure 4: Distribution of the ratings by domain and construct. Substitution is modal at 2 and complementarity at 4 in both domains, the zero category is used most for green complementarity, and the top of the scale draws at most 3 percent of the ratings of any domain and construct.

Complementarity is rated above substitution in both domains. The respondent model rates substitution about a quarter of a scale point lower than experts do (Rony and Ciarli, 2026), so the level of this gap is partly a property of the model. The analysis below relies on its variation across technologies, occupations and industries. Mean complementarity exceeds mean substitution by more than one scale point in both domains, and at the triplet level, averaging the fifteen ratings per construct, substitution is rated above complementarity in about one triplet in ten. The two constructs move together across triplets, correlating at 0.69 in the digital and 0.83 in the green domain (Figure 5). Both rise where the technology affects the occupation's work, and direction is the gap between them. Section 4.4 keeps both as levels. The personas move the level of the ratings and leave their ranking, and the zero category functions as a no-effect code (Section 6).

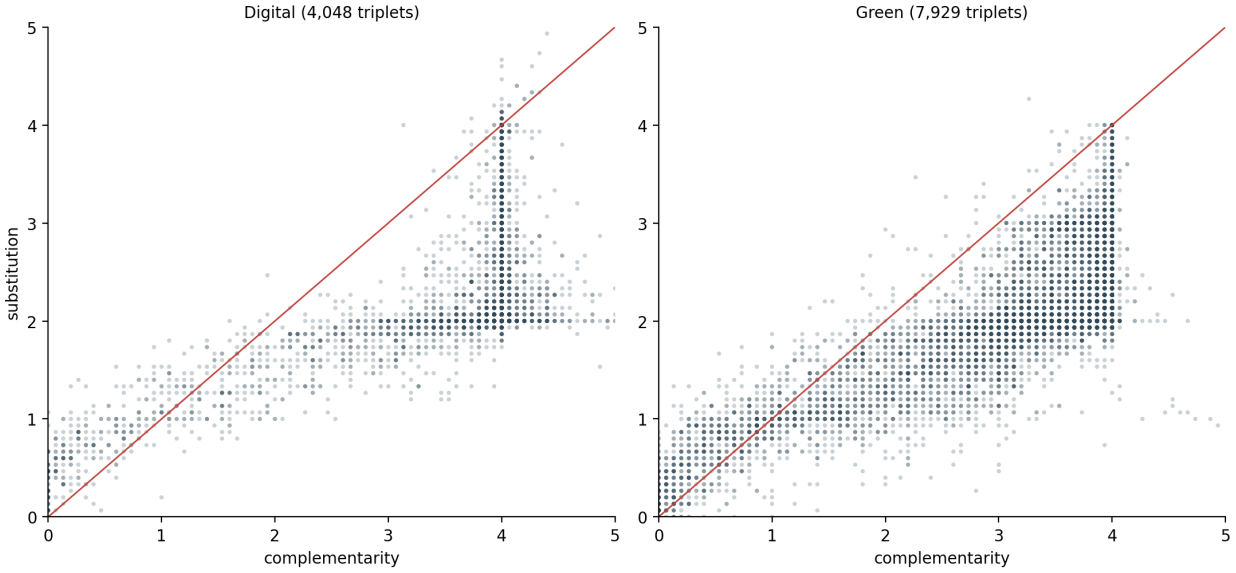


Figure 5: Triplet-level substitution against complementarity. Each point is a rated cell, and each coordinate is a mean over fifteen ratings. About 90 percent of the cells lie below the diagonal, and substitution rises with complementarity. Substitution exceeds complementarity in 354 of 4,048 digital and 795 of 7,929 green triplets.

4.4 From ratings to substitution and complementarity content

The ratings become a measure by reading each answer as a fraction of the task content the technology reaches and multiplying that fraction by the exposure. The three agreeing grades count as half, three quarters and all of the reached content, following their scale labels. The disagreeing grades and the zero count as none, so that a disagreeing answer never adds content, although the label of the higher disagreeing grade names a quarter. Section 6.5 shows the zero and the lowest grade to be the same answer. Averaged over the fifteen ratings of a triplet the mapping gives a substitution fraction $\sigma_{n,o,i}$ and a complementarity fraction $\kappa_{n,o,i}$ for technology n, occupation group o and section i. Across the triplets σ is zero in about half and averages below a fifth, and κ averages above one half and is zero in a tenth or fewer. The mapping lowers the rank correlation between the two fractions from 0.62 and 0.77 on the raw scale to about one half, and σ exceeds κ in about one triplet in a hundred.

An entity's content is its exposure to each technology times the rated fraction, summed over

technologies. For a four-digit occupation e in group $o(e)$, with $x_{en}(w)$ its exposure to technology n over the window w of the application (Equation (1)) and N the number of technology nodes of the domain,

$$\Sigma_e \;=\; \frac{1}{N}\sum_n \sum_i \omega_{o(e)i}\; x_{en}(w)\, \sigma_{n,o(e),i}, \qquad \Gamma_e \;=\; \frac{1}{N}\sum_n \sum_i \omega_{o(e)i}\; x_{en}(w)\, \kappa_{n,o(e),i}, \tag{2}$$

where $\omega_{o(e)i}$ is the pooled employment share of section i among the group's workers in the LFS, and cells outside the grid contribute nothing, so that a technology whose exposure to the pair fell below the thresholds is counted as reaching no content worth rating. The industry-side content of a NACE code is the mirror image. Its exposure to each technology is multiplied by the fraction rated for its section and averaged over the occupations employed in the section, weighted by their employment shares. Σ_e corresponds to the displacement of workers from tasks in the framework of Acemoglu and Restrepo (2019), and Γ_e to the augmentation of work that stays with workers, the distinction that Autor, Chin, Salomons, and Seegmiller (2024) draw for patents. The rating supplies the direction and the exposure the magnitude. Because the two constructs are rated separately, σ and κ are separate intensities. For robot manipulators both are high, and $\Sigma_e + \Gamma_e$ is neither the exposure nor bounded by it. The contents are also computed at the level at which exposure is measured, so they aggregate along the same paths as the exposure they accompany, to sections and industry groups on the industry side and to occupation groups on the occupation side, and can take the place of exposure in specifications that use it.

Table 4 ranks technologies by the substitution share of their assessed effect, $\bar{\sigma}_n/(\bar{\sigma}_n + \bar{\kappa}_n)$ over the technology's rated triplets. The digital ranking is led by robotics and artificial-intelligence technologies, service robotics, robot manipulators, autonomous farm equipment, symbolic AI and agent systems and language models, and ends with infrastructure and interface technologies, messaging middleware, operating-system kernels, graphical user interfaces, immersive technologies and display panels. The green ranking runs from the production-process technologies of cement, metals, chemicals and glass to energy efficiency in buildings, water conservation, health protection and conventional vehicles. The appearance of cement and glass production near the top reflects the labelling problem of Section 6.5, in which these nodes are rated as industrial processes and are not read as mitigation technologies. A share of two small fractions is unstable. Table 4 therefore reports both fractions, and glass and cement production rank high on fractions below a quarter. Across technologies the share has a standard deviation of 0.13 in both domains, and within a technology, across the sections in which it was rated, about half that or less.

Substitution and complementarity content are correlated. Across the 427 occupations the correlation is 0.67 for digital and 0.89 for green technologies, and across the 272 NACE codes it is 0.68 and 0.94. Both contents also correlate with exposure, at 0.71 to 0.88. Because the two contents enter the employment regression of Section 5.3 jointly, the green coefficients rest on little independent variation. Digital substitution content is zero for 19 percent of occupations, mostly those with no rated cell.

5 Validation of the direction measure

5.1 Stability and determinants of direction

For each triplet the direction is the difference between the complementarity and the substitution fraction, $\kappa_{n,o,i} - \sigma_{n,o,i}$. It is positive in about nine triplets in ten. When the fifteen ratings are split at random into two halves, the directions computed from each half correlate above 0.9 in both

Table 4: Technologies ranked by the substitution share of their assessed effect. The share is $\bar{\sigma}_n/(\bar{\sigma}_n + \bar{\kappa}_n)$ over the technology's rated triplets, and the table lists the top and bottom five per domain among technologies with at least twenty rated triplets.

Technology	$\bar{\sigma}_n$	$\bar{\kappa}_n$	Share	Triplets
Digital, highest				
Service Robotics	0.65	0.74	0.47	70
Robot Manipulators and Arm Systems	0.59	0.72	0.45	82
Autonomous Farm Equipment	0.23	0.29	0.44	26
Symbolic AI, Knowledge and Agent Systems	0.51	0.74	0.41	83
NLP, Generative AI and Language Models	0.51	0.76	0.40	92
Digital, lowest				
Messaging Middleware	0.00	0.68	0.00	24
OS Kernel, Process and Memory Management	0.00	0.49	0.01	32
Graphical User Interfaces and UX Design	0.02	0.85	0.02	23
AR/VR and Immersive Technologies	0.03	0.72	0.04	55
Display Panels	0.03	0.70	0.04	37
Green, highest				
Production of cement	0.16	0.22	0.43	22
Technologies in the production process for final industrial or consumer products	0.41	0.58	0.42	322
Technologies related to metal processing	0.36	0.51	0.41	124
Technologies relating to chemical industry	0.37	0.54	0.41	68
Glass production	0.07	0.12	0.38	43
Green, lowest				
Conventional vehicles (based on internal combustion engine)	0.00	0.23	0.00	224
In human health protection, e.g. against extreme weather	0.00	0.53	0.00	214
Water conservation; Efficient water supply; Efficient water use	0.01	0.59	0.01	197
Energy efficiency in buildings	0.00	0.44	0.01	409
Fuel efficiency-improving vehicle design	0.00	0.34	0.01	38

domains, a split-half stability of about 0.95 by the Spearman–Brown formula (Table 5). At the low sampling temperature the draws vary little by design, so the statistic captures the consistency of the sampled ratings and says nothing about agreement between independent raters. The rating noise in a triplet's direction has a standard deviation of 0.05. Table 5 decomposes the variance of $\kappa - \sigma$ across triplets. The technology alone accounts for the largest share in the digital domain, the technology and the occupation for similar shares in the green domain, and the industry alone for 5 percent or less. The combination of technology and occupation accounts for about four fifths of the variance, which, given the share of the variance that is stable across draws, leaves at most about a sixth for the industry in which a technology meets an occupation. Among the roughly 1,000 digital and 1,600 green technology-occupation pairs rated in two or more industries, the direction varies across industries by about twice the rating noise but changes sign beyond two noise standard deviations in one pair in five hundred or fewer. The largest differences run from net substitution or no relation in one industry to strong complementarity in another, as for electronic payment processing among keyboard operators, negative in information and communication and strongly positive in accommodation and food service activities.

Table 5: Stability and determinants of the direction $\kappa - \sigma$ across rated triplets.

	Digital	Green
A. Ratings		
Triplets	4,048	7,929
Mean direction $\kappa - \sigma$	0.459	0.381
Share of triplets with $\kappa > \sigma$	0.90	0.89
Split-half correlation	0.931	0.908
Split-half reliability (Spearman-Brown)	0.964	0.952
Standard deviation of the rating noise in $\kappa - \sigma$	0.051	0.053
B. Share of the variance of $\kappa - \sigma$ explained by fixed effects		
Technology	0.430	0.273
Occupation	0.231	0.247
Industry	0.040	0.048
Technology, occupation and industry, additive	0.585	0.496
Technology × occupation	0.812	0.780
Technology × industry	0.538	0.400
Occupation × industry	0.311	0.315
C. Technology-occupation pairs rated in two or more industries		
Pairs	999	1,586
Standard deviation of $\kappa - \sigma$ across industries (mean over pairs)	0.103	0.110
Pairs changing sign beyond two noise standard deviations (percent)	0.2	0.1

Direction is $\kappa - \sigma$ per technology-occupation-industry triplet, with each fraction the mean over fifteen ratings mapped as in Section 4.4. For the split halves, the ratings of each triplet and construct are divided at random into two halves, and the noise standard deviation is half the standard deviation of the difference between the halves' directions. Panel B reports the R^2 of $\kappa - \sigma$ on the fixed effects named. Panel C covers technology-occupation pairs rated in at least two sections, and a sign change requires $\kappa - \sigma$ above twice the noise standard deviation in one section and below its negative in another.

5.2 Aggregation and the direction signal

Table 6 reports the two contents at each level of aggregation. Across NUTS-2 regions (the EU's second-level statistical regions) their standard deviation is a quarter or less of that at any other

level, and across the ten industry groups from which regional values are built they correlate at 0.88 for digital and 0.91 for green. Regions therefore cannot separate the two contents. Across three-digit occupation groups they correlate at 0.61 for digital and 0.88 for green. Appendices D.8 and D.9 compare the contents with single indices of exposure and direction and with a usage-based index.

Table 6: The two contents by level of aggregation, with exposure over 2009–2023. Units without rated content are counted at zero. Three-digit occupation groups, NACE sections and the ten industry groups are unweighted means of their four-digit occupations or three-digit industries, and regional values weight the ten group values by each region's 2009 employment shares.

		Substitution content		Complementarity content		
Level	Units	Mean	Standard deviation	Mean	Standard deviation	Correlation of the two contents
A. Digital						
Four-digit occupations	427	0.012	0.016	0.041	0.060	0.67
Three-digit occupation groups	127	0.012	0.014	0.042	0.063	0.61
Three-digit NACE industries	272	0.010	0.012	0.034	0.062	0.68
NACE sections	21	0.006	0.008	0.028	0.051	0.80
Ten industry groups	10	0.007	0.008	0.040	0.070	0.88
NUTS-2 regions	313	0.007	0.001	0.027	0.004	0.66
B. Green						
Four-digit occupations	427	0.157	0.192	0.571	0.514	0.89
Three-digit occupation groups	127	0.147	0.190	0.545	0.503	0.88
Three-digit NACE industries	272	0.218	0.176	0.701	0.497	0.94
NACE sections	21	0.151	0.202	0.566	0.625	0.97
Ten industry groups	10	0.119	0.130	0.502	0.448	0.91
NUTS-2 regions	313	0.145	0.032	0.551	0.099	0.97

5.3 Substitution and complementarity content and employment change

For occupation group o in country c over a period from t_0 to t_1, Table 7 estimates

$$\ln E_{co,t_1} - \ln E_{co,t_0} \;=\; \beta_\Sigma \, \Sigma_o + \beta_\Gamma \, \Gamma_o + \phi \ln E_{co,t_0} + \mu_{\text{ISCO1}} + \delta_c + \varepsilon_{co}, \tag{3}$$

where $E_{co,t}$ is employment, corrected for the 2021 break in the survey series, Σ_o and Γ_o the substitution and complementarity contents of one domain, built from the patents filed in the years of the period, averaged over the four-digit codes of the group and divided by their standard deviation across cells, μ_{ISCO1} fixed effects for the ISCO major group and δ_c country fixed effects. The digital and the green pair are estimated in separate regressions, weighted by base-year employment, with standard errors clustered by occupation group, on the cells observed in all three periods. Under the displacement reading of the task framework $\beta_\Sigma < 0$, and under the augmentation reading $\beta_\Gamma > 0$. Occupation series at this detail begin in 2011, with ISCO-08, so the full period is 2011–2023, which the table also splits into 2011–2019 and 2020–2023. Columns (1) to (3) use the LFS microdata cells at three digits for 21 countries, the 24 countries of the microdata extract less Bulgaria, Slovenia and Malta, which report occupation at one or two digits. Germany, Austria and the Netherlands are absent from the extract. Columns (4) to (6) use the public Eurostat table of employment by two-digit occupation for the same 21 countries, so that the test can be reproduced without access to the microdata.

Over 2011–2023 in the microdata, one standard deviation more digital complementarity content is associated with employment growth about 0.2 log points higher and one standard deviation more digital substitution content with growth about 0.12 log points lower, both significant at 1 percent. Green substitution content is negative and not significant (Table 7). In the public table only the green substitution content is significant. Of the two sub-periods, the association appears in 2020–2023. Over 2020–2023 both digital contents carry their expected signs and are significant in both sources, with coefficients about half as large in the public table, and the green substitution content is negative in both and significant in the public table. Over 2011–2019 no substitution content is significant, and the complementarity contents are significant only at 10 percent, in one source each. Extending the public table to all member states lowers the digital estimates for 2020–2023 to −0.035 and 0.027, both still significant. Without Germany, whose occupation series carries a break flag in the Eurostat table, they are −0.043 and 0.035, close to those on the 21 countries.

The respondent model was trained on text that postdates the outcome periods and may describe the occupational changes this test measures, so the test is not out of sample for the rater. If the ratings reflected knowledge of past employment trends, the association would appear in both periods, but over 2011–2019 no substitution content is significant and the complementarity contents only at 10 percent. This does not establish that the rater was uninformed by the outcomes.

Table 7: Substitution and complementarity contents and employment change across country-by-occupation cells. Employment change is over 2011–2019, 2020–2023 and 2011–2023, in the LFS microdata at three-digit ISCO in columns (1) to (3) and in the public Eurostat table at two-digit ISCO in columns (4) to (6), and the contents are built from the patents filed in the years of each period.

	Microdata, ISCO three-digit			Public table, ISCO two-digit		
Change in log employment	2011–2019	2020–2023	2011–2023	2011–2019	2020–2023	2011–2023
	(1)	(2)	(3)	(4)	(5)	(6)
A. Digital contents						
Digital substitution content	0.009	-0.081***	-0.125***	0.055	-0.044**	-0.023
	(0.027)	(0.017)	(0.046)	(0.051)	(0.021)	(0.059)
Digital complementarity content	0.055*	0.067***	0.196***	0.010	0.033**	0.064
	(0.033)	(0.013)	(0.048)	(0.036)	(0.015)	(0.043)
B. Green contents						
Green substitution content	0.022	-0.046	-0.088	-0.038	-0.047**	-0.088**
	(0.031)	(0.029)	(0.066)	(0.028)	(0.020)	(0.037)
Green complementarity content	0.003	0.007	0.067	0.061*	-0.001	0.068
	(0.039)	(0.036)	(0.068)	(0.034)	(0.029)	(0.045)
ISCO major-group fixed effects	✓	✓	✓	✓	✓	✓
Country fixed effects	✓	✓	✓	✓	✓	✓
Log base-year employment	✓	✓	✓	✓	✓	✓
Base-year employment weights	✓	✓	✓	✓	✓	✓
Occupation clusters	120	120	120	40	40	40
Observations	1,739	1,739	1,739	786	786	786

Change in log employment of country-by-occupation cells observed in all three periods, on the contents built from the patents filed in the years of the same period, averaged over the four-digit codes of the occupation group and divided by their standard deviation. The digital and the green pair are estimated in separate regressions. Columns (1) to (3) use the EU Labour Force Survey microdata and columns (4) to (6) the public Eurostat table of employment by two-digit occupation, for the same 21 EU countries. Occupation series begin in 2011, with ISCO-08. Standard errors clustered by occupation group are in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

6 Design choices and further results of the direction measure

6.1 The rating prompt

Each call sends a system message, which sets the model's standing instructions, and a user message, which poses the question. The placeholder `{domain}` is digital or green and the bracketed fields are filled from the triplet. The system message is reproduced below.

```
You are an expert in {domain} technologies. Your expertise is defined by the
given persona and context in user prompt.
Task: Given a prompt, produce exactly ONE JSON object evaluating the statement.

STRICT OUTPUT FORMAT:
- Output MUST be a single JSON object with exactly these keys in this order:
  1) "rating": an integer from 0 to 5 (no decimals).
  2) "reason": a 1-3 lines string (<= 25 words), plain text.
- No additional keys, no trailing commas.
- No prose before or after the JSON. No code fences. No comments. No markdown.

POLICY:
- If information is ambiguous, choose the closest rating and explain briefly.
- Never ask questions or request more info.
- Use a neutral, expert tone; avoid hedging unless required by the persona.
- 0 means you do not know. Never use 0 to signal a refusal.
```

The user message is reproduced below.

```
You are assessing how a {domain} technology will affect an occupation as it is
practised within a specific industry.
- Substitute: the {domain} technology takes over the work itself, so fewer
  human workers are needed to do it.
- Complement: the {domain} technology supports or enhances the human workers,
  who still do the work, but better, faster or at higher quality.

You are {role}. Answer strictly from that perspective.

Occupation: [occupation group title]
Industry: [NACE section title]
{Domain} technology: [technology name or ENV-TECH description]

To what extent do you agree with the statement below?
- 1 = Strongly disagree
- 2 = Disagree (25 percent)
- 3 = Neutral / somewhat agree (50 percent)
- 4 = Mostly agree (75 percent)
- 5 = Fully agree
- 0 = Do not know

Statement: {statement}

Provide exactly one rating from [0, 1, 2, 3, 4, 5] and one concise reason
(1-3 sentences).
```

The role is "a manager who runs the team that performs this work day to day", "an experienced worker who personally performs this task day to day" or "a chief executive officer responsible for the

organisation's strategy and investment decisions". The substitution statement is "This {domain} technology will replace human workers in this occupation within this industry." and the complementarity statement "This {domain} technology will work alongside human workers in this occupation within this industry and support them." The response is constrained by a JSON schema with an integer rating from 0 to 5 and a string reason.

6.2 The prompt-format experiment

Whether consolidated prompt formats lose information was tested on twelve triplets drawn from the exposure grid, one central and one peripheral triplet for each of the three highest-exposure technologies of each domain. Each of four formats with identical wording was run with five draws, for 720 calls and 1,440 ratings in total. The four formats are the separated format, with one construct and one persona per call, a format merging the two constructs, a format merging the three personas, and a fully merged format. The experiment was run before the production prompt was fixed, at temperature and top-p of 1.0, with a horizon of 2030, a human-resources manager in place of the chief executive and the zero defined as do not know or no effect, on triplets from an earlier exposure build. The comparison holds the prompt fixed across formats. Table 8 reports it. Agreement with the separated reference declines monotonically with consolidation, and individual cell means shift by up to three scale points. Use of the no-effect category falls when personas share a call, from 7.8 to 0.3 percent overall with merged personas, and on the single no-effect triplet in the test set from 36.7 to 3.3 percent. The latter comparison rests on one triplet and shows the sign of the change without estimating its size. Draw-to-draw variability increases monotonically with consolidation, and the merged-construct format widens the substitution-complementarity gap by half a scale point, which would bias the measured direction directly. The separated format was therefore retained despite its sixfold cost.

Table 8: Prompt-format experiment on twelve triplets, with five draws per format. The separated format is the reference.

	Separated	Merged constructs	Merged personas	Fully merged
Calls per triplet-draw	6	3	2	1
Correlation with separated cell means		0.883	0.805	0.764
Mean absolute deviation		0.42	0.52	0.53
Zero share, all ratings (%)	7.8	4.2	0.3	3.1
Zero share, no-effect triplet (%)	36.7	33.3	3.3	10.0
Within-prompt draw SD	0.475	0.523	0.587	0.646
Net rating (substitution minus complementarity), mean	−1.056	−1.572	−0.811	−1.146

6.3 The grid design space

The grid is one point in a design space with six dimensions. These are occupation resolution (three- or four-digit ISCO), industry resolution (NACE section, division or group), technology resolution (base nodes, intermediate levels, or one indicator per domain), the exposure filter (none, top-K per margin, or percentile cuts), the source recording occupation-industry pairs (the LFS, the OECD Survey of Adult Skills, online vacancies, or none), and the co-occurrence floor. The adopted design combines the finest resolution at which a representative EU-wide employment source records pairs, three-digit occupations by sections under the LFS, with the common exposure thresholds

of Section 4.2, which are 1.31 for digital occupations, 1.78 for digital industries, 3.79 for green occupations and 4.15 for green industries. The 0.1 percent employment floor is the co-occurrence cut in the LFS, where presence alone admits pairs with negligible employment. Four-digit occupations could be recorded only through the Survey of Adult Skills (PIAAC), which covers thirteen EU countries and in which any share-based floor reduces to requiring that the pair be observed at all, because the median occupation counts about 140 respondents. The experiment log reports the sizes of the alternative grids.

6.4 Personas

The personas move the level of the ratings and leave their ranking. The chief executive returns the highest mean substitution rating in both domains, 2.38 in the digital and 2.06 in the green, and the worker the lowest, 2.10 and 1.85, while complementarity differs little across roles, from 3.50 to 3.43 in the digital and from 3.07 to 2.88 in the green domain (Figure 6). Across triplets the three persona means correlate between 0.88 and 0.95 and differ in absolute value by 0.17 to 0.35 scale points. In a decomposition of the variance of the individual ratings, between 70 and 84 percent lies between triplets, 4 to 7 percent between the personas of one triplet, and 12 to 23 percent between draws of one persona.

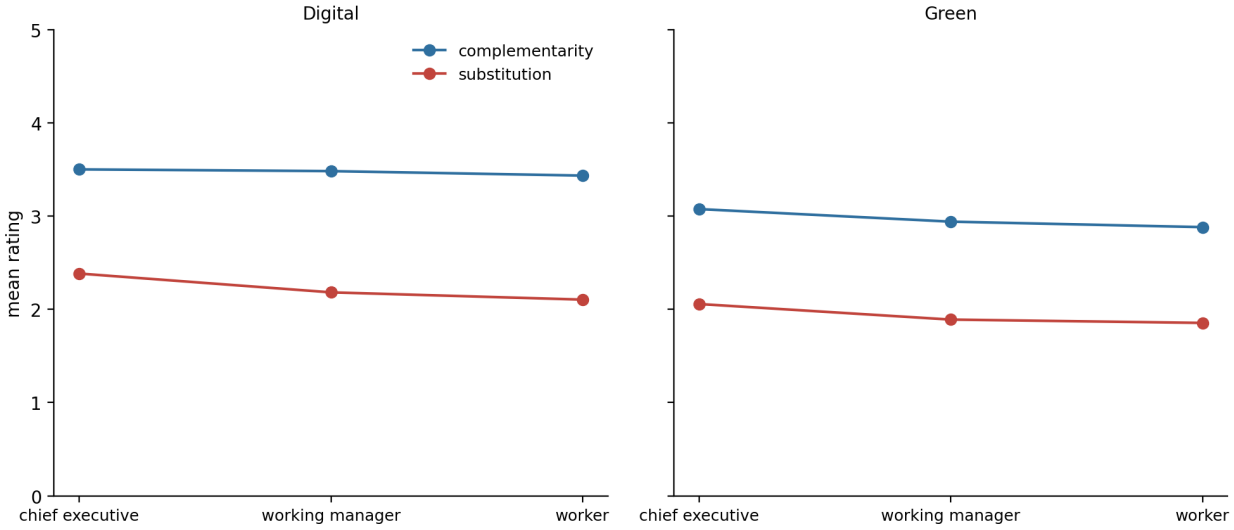


Figure 6: Mean rating by persona, construct and domain. The three roles differ by at most 0.28 scale points on substitution and 0.19 on complementarity, with the chief executive highest on substitution and the worker lowest in both domains.

6.5 How the zero category is used

Zeros are 9.4 percent of all ratings, 7.4 and 3.4 percent for digital complementarity and substitution and 13.4 and 9.6 percent for their green counterparts, and the worker uses the category almost twice as often as the chief executive, in 12.1 against 6.9 percent of answers. Although the scale defines the zero as do not know, the justifications show that it functions as a no-effect code. Outside the twelve green nodes discussed below, which account for 30.8 percent of the zeros, the stated reason is typically that the technology has no relation to the work of that occupation in that industry, which is the content of a rating of 1. The two categories are interchangeable at the level of the individual

response. The same persona rating the same triplet returns the same justification as a zero in one draw and as a 1 in another. For additive manufacturing with financial and mathematical associate professionals in accommodation and food service, the worker's zero reads "I see no direct connection between 3D printing and my daily tasks like budgeting or data analysis" and the worker's 1 "I see no direct link between 3D printing and my core tasks like budgeting or revenue analysis. It does not support or enhance my daily accounting or reporting work." A zero is therefore retained as a zero, and S and C are means over all fifteen ratings per construct on a scale running from 0 to 5, with the zero as the bottom category.

The exception is a set of twelve green nodes whose ENV-TECH descriptions are not readable as technologies once detached from their parent heading. Six name an activity that is not itself green, among them conventional vehicles, aeronautics or air transport, glass production and nuclear fission reactors, and six are umbrella descriptions such as enabling technologies with a potential contribution to emissions mitigation or technologies having an indirect contribution to adaptation. For these nodes the zero records that the label cannot be rated and does not mean that the technology leaves the work unaffected. Their zeros are 1.7 and 1.2 percent of all ratings respectively, 2.9 percent together, while the twelve nodes carry 19 percent of all ratings. Section 6.6 lists them with their zero shares and the justifications on which the reading rests.

6.6 Green nodes not readable as technologies

The justifications of the zero ratings separate the twelve nodes into two groups. In the first, the ENV-TECH description names an activity that is not green once separated from its parent heading on climate change mitigation, and the respondent rejects the premise of the question. For glass production rated for refuse workers in water supply and waste management, one justification reads "Glass production is not a green technology used in waste management; it is a separate manufacturing process." In the second, the description is an umbrella formulation that names no technology. For technologies with an indirect contribution to adaptation rated for drivers in manufacturing, one justification reads "The technology is too vaguely defined to assess its direct impact on driving tasks in manufacturing." Table 9 reports the twelve nodes. The nodes are retained with the ratings as given, and the reading of the zero adopted in Section 4.3 applies to the zeros outside these nodes, about 69 percent of all zeros.

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Table 9: The twelve green nodes whose descriptions are not readable as technologies, with their use of the zero category. Together they carry 30.8 percent of all zero ratings, and their zeros are 2.9 percent of all ratings.

Node description	Triplets	Ratings	Zeros	Share (%)
Activity that is not green without its parent heading				
Conventional vehicles (based on internal combustion engine)	224	6,720	1,775	26.4
Aeronautics or air transport	107	3,210	1,491	46.4
Glass production	43	1,290	883	68.4
Production of cement	22	660	212	32.1
Nuclear fission reactors	85	2,550	1,448	56.8
Ceramic production	23	690	363	52.6
Umbrella description naming no technology				
Technologies having an indirect contribution to adaptation to climate change	379	11,370	1,723	15.2
Enabling technologies with a potential contribution to GHG emissions mitigation	812	24,360	730	3.0
At coastal zones; at river basins	60	1,800	930	51.7
Enabling technologies or technologies with a potential or indirect contribution to GHG emissions mitigation	132	3,960	193	4.9
Technologies in the production process for final industrial or consumer products	322	9,660	587	6.1
Climate change mitigation technologies for sector-wide applications	88	2,640	91	3.4

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